Social Web Analytics - Group Project

Question 1

```
library(RedditExtractoR)
library(tidyverse)
```

a. Using the Reddit API, identify relevant thread URLs for your chosen topic. Focus on a monthly or yearly timeframe to ensure sufficient data depth.

Since the API requests sometimes get error messages due to limited access, I find the threads through API and save it to a file for later use without repeated execution.

```
# Note: This file is executed only once to avoid error
# Find all threads having "AI" keyword in the ArtificialInteligence community
threads <- find_thread_urls(
    subreddit = "ArtificialInteligence",
    sort_by = "top",
    period = "year")

# Save threads to another file for later opening
saveRDS(threads, "raw_reddit_data.rds")
```

```
#import and read the thread data
raw_threads <- readRDS("raw_reddit_data.rds")
glimpse(raw_threads)
```

```
Rows: 497
Columns: 7
$ date_utc  <chr> "2025-12-11", "2025-11-29", "2025-10-15", "2025-09-10", "202~
```

```
$ timestamp <dbl> 1765480956, 1764408616, 1760498359, 1757529811, 1746965679, ~
$ title     <chr> "Something Ominous Is Happening in the AI Economy", "How com~
$ text      <chr> "If the AI revolution fails to materialize as expected, the ~
$ subreddit <chr> "ArtificialInteligence", "ArtificialInteligence", "Artificia~
$ comments  <dbl> 198, 87, 50, 697, 399, 60, 118, 112, 460, 196, 19, 71, 15, 3~
$ url       <chr> "https://www.reddit.com/r/ArtificialInteligence/comments/1pk~
```

Get the Upvote scores of posts with each corresponding URL

```r
# Note: This code block requires a long time of execution
urls <- raw_threads$url
meta_rw_thread <- list()

for (i in 1:length(urls)) {
  message(paste("Processing thread number", i, "at:", urls[i]))

  # Use tryCatch (skip broken links)
  result <- tryCatch({
    # Get thread content
    get_thread_content(urls[i])
  }, error = function(e) {
    # If Error -> Print Message -> Return NULL
    message(paste("Url Error at the thread number", i, ":", e$message))
    return(NULL)
  })

  # Save result whether it's null or valid
  meta_rw_thread[[i]] <- result

  # Short pausing time
  Sys.sleep(3)
}

# Remove all Null data, only keep valid URL
meta_thread_cleaned <- meta_rw_thread[!sapply(meta_rw_thread, is.null)]
```

```r
# Save threads to another file for later opening
saveRDS(meta_thread_cleaned, "meta_thread_cleaned.rds")
```

```r
# Read metadata of threads collected
meta_threads <- readRDS("meta_thread_cleaned.rds")
```

b. Find the top three threads with the highest Score (Upvotes). Retrieve and display the title and main post body of each of these three threads.

```
# COMBINE ALL THREADS' METADATA INTO A SINGLE DATAFRAME
library(purrr)

# 1/ Combine all the thread metadata of each post on
# Reddit into a whole dataframe
df_only_threads <- map_dfr(meta_threads, ~ .x$threads)

# 2/ Take the index number of the Top 3 highest upvotes
top_3_indices <- df_only_threads |>
  # set the index number as an original order in df
  mutate(origin_index = row_number()) |>
  arrange(desc(upvotes)) |> # order by the upvotes descendingly
  slice(1:3) |> # top 3 only
  pull(origin_index) # only keep the index number

# 3.1/ Top 3 thread
df_top3_thread <- meta_threads[top_3_indices] |>
  map_dfr(~ .x$threads)

# 3.2/ Top 3 threads' comments
df_top3_comment <- meta_threads[top_3_indices] |>
  map_dfr(~ .x$comments)
```

```
library(stringi)
library(stringr)

# Convert strings to UTF-8
# and fix/remove invalid byte sequences to prevent processing errors.

df_top3_thread$text <- stri_enc_toutf8(df_top3_thread$text, validate = TRUE)
df_top3_thread$title <- stri_enc_toutf8(df_top3_thread$title, validate = TRUE)

# Use Regular Expressions to strip non-ASCII characters
# (emojis, symbols, and special icons).
df_top3_thread$text <- str_replace_all(df_top3_thread$text,
                                       "[^\\x20-\\x7E]", "")
df_top3_thread$title <- str_replace_all(df_top3_thread$title,
                                        "[^\\x20-\\x7E]", "")
```

```r
# Similar process for the comment content
df_top3_comment$comment <- stri_enc_toutf8(df_top3_comment$comment,
                                            validate = TRUE)
df_top3_comment$comment <- str_replace_all(df_top3_comment$comment,
                                           "[^\\x20-\\x7E]", "")

# Observe required dataframes
glimpse(df_top3_thread) # These are the top 3 thread with highest upvotes
```

```
Rows: 3
Columns: 15
$ url                   <chr> "https://www.reddit.com/r/ArtificialInteligence/~
$ author                <chr> "shallow-pedantic", "reddit20305", "cheesomaciti~
$ date                  <chr> "2025-10-22", "2025-11-08", "2025-04-30"
$ timestamp             <dbl> 1761157898, 1762572350, 1745988845
$ title                 <chr> "I was once an AI true believer. Now I think the~
$ text                  <chr> "I used to be all-in on large language models. B~
$ subreddit             <chr> "ArtificialInteligence", "ArtificialInteligence"~
$ score                 <dbl> 6252, 5603, 3899
$ upvotes               <dbl> 6252, 5603, 3899
$ downvotes             <dbl> 0, 0, 0
$ up_ratio              <dbl> 0.95, 0.97, 0.96
$ total_awards_received <dbl> 0, 0, 0
$ golds                 <dbl> 0, 0, 0
$ cross_posts           <dbl> 6, 6, 9
$ comments              <dbl> 1245, 681, 1918
```

```r
glimpse(df_top3_comment) # These are the comments of the top 3
```

```
Rows: 1,415
Columns: 10
$ url        <chr> "https://www.reddit.com/r/ArtificialInteligence/comments/1o~
$ author     <chr> "AutoModerator", "Baffin622", "steveb858", "PieOhMy69420", ~
$ date       <chr> "2025-10-22", "2025-10-22", "2025-10-22", "2025-10-23", "20~
$ timestamp  <dbl> 1761157899, 1761158234, 1761160658, 1761182798, 1761194796,~
$ score      <dbl> 1, 1056, 197, 129, 96, 36, 27, 3, 2, 2, 2, 17, 24, 2, 2, 9,~
$ upvotes    <dbl> 1, 1056, 197, 129, 96, 36, 27, 3, 2, 2, 2, 17, 24, 2, 2, 9,~
$ downvotes  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ golds      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ comment    <chr> "## Welcome to the r/ArtificialIntelligence gateway### Ques~
$ comment_id <chr> "1", "2", "2_1", "2_1_1", "2_1_1_1", "2_1_1_1_1", "2_1_1_1_~
```

```r
# print DATE, TITLE, and BODY of those threads

for (i in 1:3) {
    cat("\n\nThread", i, "\n") # Header
    cat("---", "\n")
    cat("DATE:", df_top3_thread$date[i], "\n")
    cat("UPVOTE:", df_top3_thread$upvotes[i], "\n\n")

    # Wrap Title (allow 80 strings per line)
    clean_title <- str_wrap(df_top3_thread$title[i], width = 80)
    cat("TITLE:\n", clean_title, "\n\n")

    # Wrap Body/Text (allow 80 strings per line)
    # str_wrap to automatically wrap the body
    clean_text <- str_wrap(df_top3_thread$text[i], width = 80)
    cat("BODY:\n", clean_text, "\n")

    cat("\n============================\n")
}
```

```
Thread 1
---
DATE: 2025-10-22
UPVOTE: 6252

TITLE:
 I was once an AI true believer. Now I think the whole thing is rotting from the
inside.

BODY:
 I used to be all-in on large language models. Built automations, client tools,
business workflows..... hell, entire processes around GPT and similar systems.
I thought we were seeing the dawn of a new era. I was wrong.Nothing is reliable.
If your workflow needs any real accuracy, consistency, or reproducibility,
these models are a liability. Ask the same question twice and get two different
answers. Small updates silently break entire chains of logic. Its like building
on quicksand.That old line, *this is the worst itll ever be,* is bullshit.
GPT-4.1 workflows that ran perfectly are now useless on GPT-5. Things regress,
behaviors shift, context windows hallucinate. You cant version-lock intelligence
that doesnt actually understand what its doing.The time and money that go into
```

guardrailing, safety layers, and compliance dwarfs just paying a human to do the work correctly. Worse, the safeguards rarely even function. You end up debugging an AI that wont admit its wrong, wrapped in another AI that cant explain why.And then theres the hype machine. Every company is tripping over itself to bolt AI-powered onto products that dont need it. Copilot, ChatGPT, Geminitheyre all mediocre at best, and big tech is starting to realize it. Real productivity gains are vanishingly rare. The MASSIVE reluctance of the business world to say something is simply due to embarrassment of admission. CEO's are literally scrambling to re-hire, or pay people like ME to come in and fix some truly horrific situations. (I am too busy fixing all of the broken shit on my end to even think about having the time to do this for others. But the phone calls and emails are piling up. Other consultants I speak with say the same thing. Copilot easily being the most requested to be fixed).Random, unreliable, and broken systems with zero audit requirements in the US. And I mean ZERO accountability. The amount of plausible deniability massive companies have to purposely or inadvertently harm people is overwhelming. These systems now influence hiring, pay, healthcare, credit, and legal outcomes without auditability, transparency, or regulation. I work with these tools every day, and have from jump. I am confident we are at minimum in a largely stalled performance drought, and at worst, witnessing the absolute floors starting to crumble.

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Thread 2

DATE: 2025-11-08
UPVOTE: 5603

TITLE:
Meta just lost $200 billion in one week. Zuckerberg spent 3 hours trying to explain what they're building with AI. Nobody bought it.

BODY:
So last week Meta reported earnings. Beat expectations on basically everything. Revenue up 26%. $20 billion in profit for the quarter but Stock should've gone up right? Instead it tanked. Dropped 12% in two days. Lost over $200 billion in market value. Worst drop since 2022.Why? Because Mark Zuckerberg announced they're spending way more on AI than anyone expected. And when investors asked what they're actually getting for all that money he couldn't give them a straight answer.**The spending**: Meta raised their 2025 capital expenditure forecast to $70-72 billion. That's just this year. Then Zuckerberg said next year will be "notably larger." Didn't give a number. Just notably larger.

Reports came out saying Meta's planning $600 billion in AI infrastructure spending over the next three years. For context that's more than the GDP of most countries. Operating expenses jumped $7 billion year over year. Nearly $20 billion in capital expense. All going to AI talent and infrastructure.During the earnings call investors kept asking the same question. What are you building? When will it make money? Zuckerberg's answer was basically "trust me bro we need the compute for superintelligence."He said "The right thing to do is to try to accelerate this to make sure that we have the compute that we need both for the AI research and new things that we're doing."Investors pressed harder. Give us specifics. What products? What revenue?His response: "We're building truly frontier models with novel capabilities. There will be many new products in different content formats. There are also business versions. This is just a massive latent opportunity." Then he added "there will be more to share in the coming months."That's it. Coming months. Trust the process. The market said no thanks and dumped the stock.Other companies are spending big on AI too. Google raised their capex forecast to $91-93 billion. Microsoft said spending will keep growing. But their stocks didn't crash. Why Because they can explain what they're getting.* Microsoft has Azure. Their cloud business is growing because enterprises are paying them to use AI tools. Clear revenue. Clear product. Clear path to profit.* Google has search. AI is already integrated into their ads and recommendations. Making them money right now.* Nvidia sells the chips everyone's buying. Direct revenue from AI boom.* OpenAI is spending crazy amounts but they're also pulling in $20 billion a year in revenue from ChatGPT which has 300 million weekly users.Meta? They don't have any of that.98% of Meta's revenue still comes from ads on Facebook Instagram and WhatsApp. Same as it's always been. They're spending tens of billions on AI but can't point to a single product that's generating meaningful revenue from it.The Metaverse dj vu is that This is feeling like 2021-2022 all over again.Back then Zuckerberg bet everything on the Metaverse. Changed the company name from Facebook to Meta. Spent $36 billion on Reality Labs over three years. Stock crashed 77% from peak to bottom. Lost over $600 billion in market value.Why? Because he was spending massive amounts on a vision that wasn't making money and investors couldn't see when it would.Now it's happening again. Except this time it's AI instead of VR.What Meta's actually building?During the call Zuckerberg kept mentioning their "Superintelligence team." Four months ago he restructured Meta's AI division. Created a new group focused on building superintelligence. That's AI smarter than humans.* He hired Alexandr Wang from Scale AI to lead it. Paid $14.3 billion to bring him in.* They're building two massive data centers. Each one uses as much electricity as a small city.But when analysts asked what products will come out of all this Zuckerberg just said "we'll share more in coming months."He mentioned Meta AI their ChatGPT competitor. Mentioned something called Vibes. Hinted at "business AI" products.But nothing concrete. No launch dates. No revenue projections. Just vague promises.The

only thing he could point to was AI making their current ad business slightly better. More engagement on Facebook and Instagram. 14% higher ad prices.That's nice but it doesn't justify spending $70 billion this year and way more next year.**Here's the issue -**Zuckerberg's betting on superintelligence arriving soon. He said during the call "if superintelligence arrives sooner we will be ideally positioned for a generational paradigm shift." But what if it doesn't? What if it takes longer?His answer: "If it takes longer then we'll use the extra compute to accelerate our core business which continues to be able to profitably use much more compute than we've been able to throw at it."So the backup plan is just make ads better. That's it.You're spending $600 billion over three years and the contingency is maybe your ad targeting gets 20% more efficient.Investors looked at that math and said this doesn't add up.So what's Meta actually buying with all this cash?* Nvidia chips. Tons of them. H100s and the new Blackwell chips cost $30-40k each. Meta's buying hundreds of thousands.* Data centers. Building out massive facilities to house all those chips. Power. Cooling. Infrastructure.* Talent. Paying top AI researchers and engineers. Competing with OpenAI Google and Anthropic for the same people.And here's the kicker. A lot of that money is going to other big tech companies.* They rent cloud capacity from AWS Google Cloud and Azure when they need extra compute. So Meta's paying Amazon Google and Microsoft.* They buy chips from Nvidia. Software from other vendors. Infrastructure from construction companies.It's the same circular spending problem we talked about before. These companies are passing money back and forth while claiming it's economic growth.**The comparison that hurts**\- Sam Altman can justify OpenAI's massive spending because ChatGPT is growing like crazy. 300 million weekly users. $20 billion annual revenue. Satya Nadella can justify Microsoft's spending because Azure is growing. Enterprise customers paying for AI tools.What can Zuckerberg point to? Facebook and Instagram users engaging slightly more because of AI recommendations. That's it.During the call he said "it's pretty early but I think we're seeing the returns in the core business."Investors heard "pretty early" and bailed.**Why this matters :**Meta is one of the Magnificent 7 stocks that make up 37% of the S&P 500. When Meta loses $200 billion in market value that drags down the entire index. Your 401k probably felt it.And this isn't just about Meta. It's a warning shot for all the AI spending happening right now.If Wall Street starts questioning whether these massive AI investments will actually pay off we could see a broader sell-off. Microsoft, Amazon, Alphabet all spending similar amounts. If Meta can't justify it what makes their spending different?The answer better be really good or this becomes a pattern.**TLDR**Meta reported strong Q3 earnings. Revenue up 26% $20 billion profit. Then announced they're spending $70-72 billion on AI in 2025 and "notably larger" in 2026. Reports say $600 billion over three years. Zuckerberg couldn't explain what products they're building or when they'll make money. Said they need compute for "superintelligence" and there will be "more to share in coming months." Stock

```
crashed 12% lost $200 billion in market value. Worst drop since 2022. Investors
comparing it to 2021-2022 metaverse disaster when Meta spent $36B and stock
lost 77%. 98% of revenue still comes from ads. No enterprise business like
Microsoft Azure or Google Cloud. Only AI product is making current ads slightly
better. One analyst said it mirrors metaverse spending with unknown revenue
opportunity. Meta's betting everything on superintelligence arriving soon. If
it doesn't backup plan is just better ad targeting. Wall Street not buying it
anymore.Sources:[https://techcrunch.com/2025/11/02/meta-has-an-ai-product-
problem/](https://techcrunch.com/2025/11/02/meta-has-an-ai-product-problem/)

============================


Thread 3
---
DATE: 2025-04-30
UPVOTE: 3899

TITLE:
 I lost my business to AI. Who else so far?

BODY:
 I ran a successful Spanish to English translation business from 2005-2023, with
5-10 subcontractors at a time and sometimes pulling 90 hour weeks and $100k+
yearly income. Now there is almost no work left because AI &amp; LLMs have
gotten so good. What other jobs have been lost? Im curious to hear your story of
losing your career to AI, if only to commiserate together.

============================
```

c. Instead of individual words, context is key. Generate a Bigram (two-word sequence) frequency chart for the comments and replies within each of these threads. Display the top 15 most frequent bigrams for each thread.

Prapare and generate a Bigram frequency chart

```r
# Observe the top 3 url of each thread
unique(df_top3_comment$url)
```

```
[1] "https://www.reddit.com/r/ArtificialInteligence/comments/1odgfys/i_was_once_an_ai_true_be
```

```
[2] "https://www.reddit.com/r/ArtificialInteligence/comments/1orewim/meta_just_lost_200_billi
[3] "https://www.reddit.com/r/ArtificialInteligence/comments/1kb8e09/i_lost_my_business_to_a
```

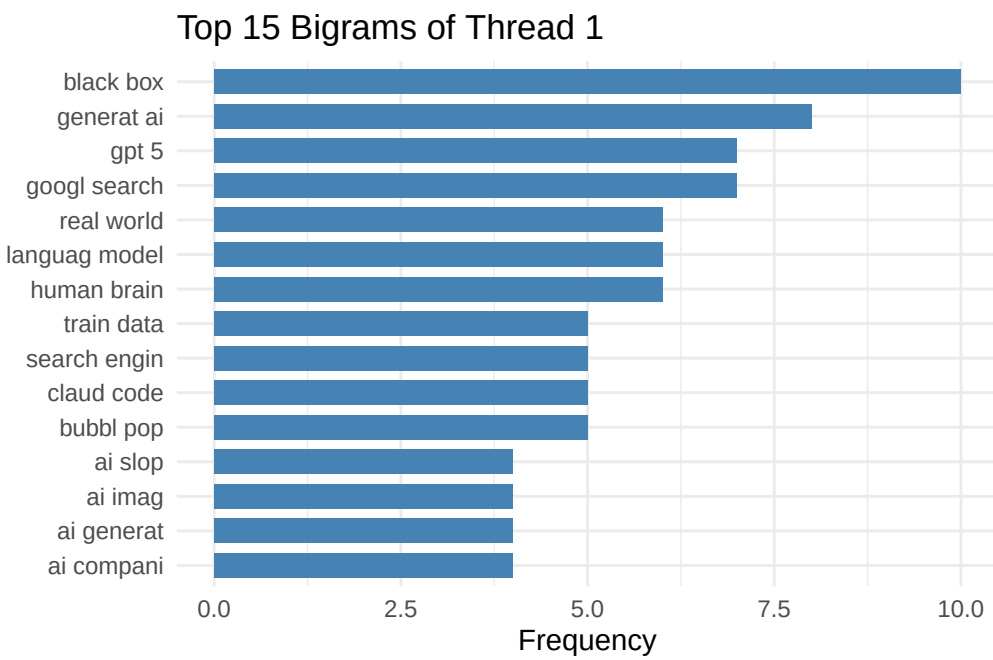
```
library(tidytext)

# 1/ Separate the comments into pairs of words
bigrams_data <- df_top3_comment |>
    select(url, comment) |> # take url as identification code of each thread
    filter(!is.na(comment), comment != "") |>
    mutate(
      comment = str_replace_all(comment, "(http\\S+|www\\.\\S+)",""), # remove link in the co
      comment = str_to_lower(comment),
      thread_num = case_when( # Add another col that mark different urls
        url == "https://www.reddit.com/r/ArtificialInteligence/comments/1odgfys/i_was_once_a
        url == "https://www.reddit.com/r/ArtificialInteligence/comments/1orewim/meta_just_los
        url == "https://www.reddit.com/r/ArtificialInteligence/comments/1kb8e09/i_lost_my_bus
        TRUE ~ "Other threads" # If all cases are not matched
      )
    ) |>
    unnest_tokens(bigram_col, comment, token = "ngrams", n = 2) |> # add bigram_col column ha
    filter(bigram_col != "")
```

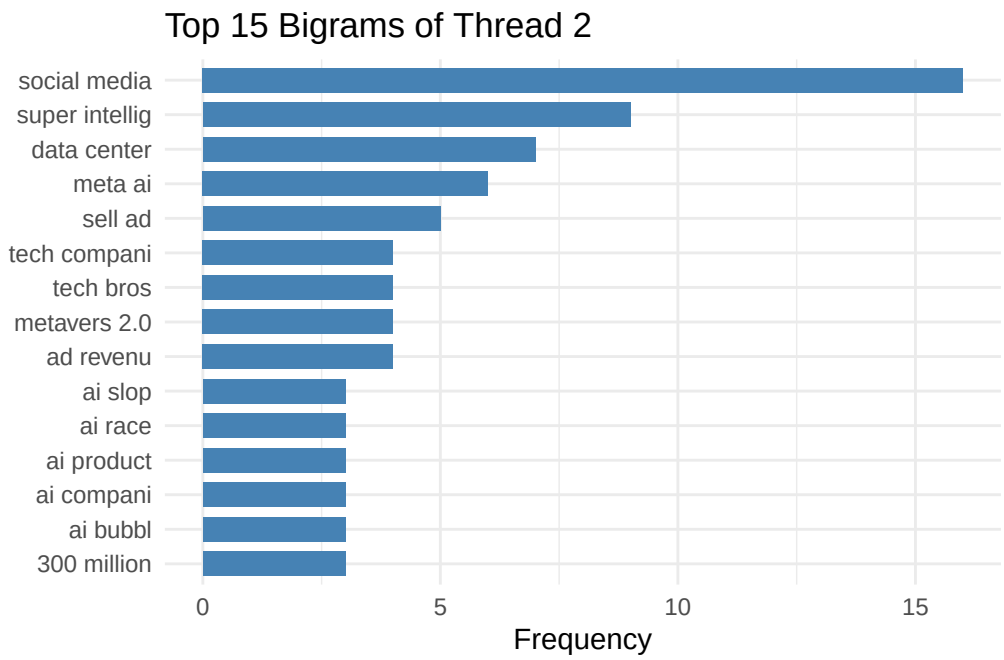
```
library(SnowballC)

# 2/ Separate pairs of words into different columns to filter stop words
bigrams_separated <- bigrams_data |>
    separate(bigram_col, c("word1","word2"), sep = " ")

# fitler rows having stop words
bigrams_filtered <- bigrams_separated |>
    filter(!word1 %in% stop_words$word) |>
    filter(!word2 %in% stop_words$word) |>
    mutate( # stemming words
      word1 = wordStem(word1, language = "en"),
      word2 = wordStem(word2, language = "en")
    )
```

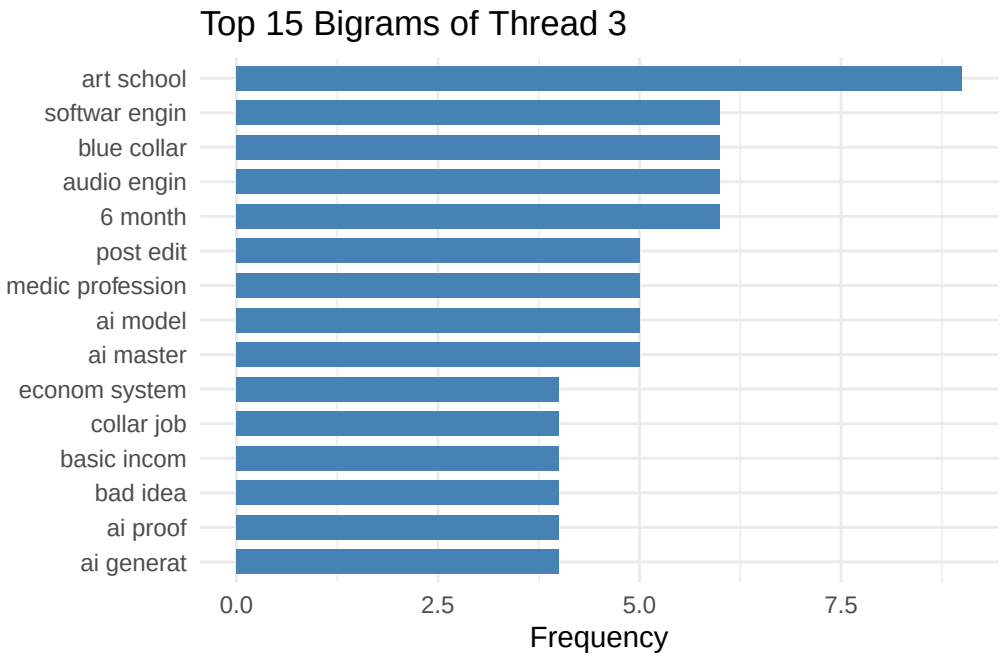
```
# 3/ Combine all word pairs and filter the top 15 of each thread
bigram_top15 <- bigrams_filtered |>
    unite(bigram_pairs, word1, word2, sep = " ") |>
    group_by(thread_num) |> # filter by identification code (each thread)
    count(bigram_pairs, sort = TRUE) |>
```

```
    # slice (select) the first 15 rows (max) of the "n" column
    slice_max(n, n=15, with_ties = FALSE) |>
    ungroup()
```

```
# 4/ Create a bigram chart for each thread
for (i in 1:3){
  current_thread <- paste0("Thread ",i)
  # Create a chart for each thread number
  thread_chart <- bigram_top15 |>
    filter(thread_num == current_thread) |>
    ggplot(aes(x = reorder(bigram_pairs,n), y = n)) +
      geom_col(fill = "steelblue", show.legend = FALSE, width = 0.7) +
      coord_flip() +
      theme_minimal() +
      labs(title= paste0("Top 15 Bigrams of ",
            current_thread),
            x = "",
            y = "Frequency")
  print(thread_chart)
}
```




Top 15 Bigrams of Thread 2
social media
super intellig
data center
meta ai
sell ad
tech compani
tech bros
metavers 2.0
ad revenu
ai slop
ai race
ai product
ai compani
ai bubbl
300 million
0
5
10
15
Frequency


Top 15 Bigrams of Thread 3
art school
softwar engin
blue collar
audio engin
6 month
post edit
medic profession
ai model
ai master
econom system
collar job
basic incom
bad idea
ai proof
ai generat
0.0
2.5
5.0
7.5
Frequency

d. Comment on your Bigram charts. Do the frequent phrases indicate a positive, negative, or neutral sentiment in the discussion? Compare the tone between the three threads.

Thread 1

In general, this thread has both negative and neutral sentiment in the discussion. While some words such as "black box", "ai slop", and "bubbl pop" may imply the skeptical and cautious comments of users related to leveraging AI in work, "claude code", "gpt 5", or "train data" have a neutral tone when they are technical terms in AI discussions.

Thread 2

Compared to the first and final threads, this post has the significantly negative sentiment. Terms such as "metaverse 2.0", "ai race", "ai bubble", or "ai slop" might convey the highly competitive market of the AI industry and the overrated AI potential on social media.

Thread 3

Similarly to the first two threads, this post has negative sentiment, which concentrates more on the social and economic aspects related to the AI development. Some words such as "blue collar", "basic income", "collar job", or "bad idea" reflect a quite anxiety of users about the future when AI could replace humans in certain jobs, thereby impacting the social structure. In addition, there are also some neutral-tone words such as "art school", "software engineer" that might mention fields affected by the development of AI.

Conclusion

In summary, high-scoring threads regarding Artificial Intelligence on Reddit tend to have negative and neutral sentiments. In addition, it can be further suggested that sharing negative feelings about a problem (in this case, AI-related stories) tends to receive more empathy and engagement from users within the community, thereby raising the upvotes (scores) of a thread.

Question 2

This is the df containing all comments in the top 15 threads

```r
# observe the structure of comment metadata of the top threads
glimpse(df_top3_comment)
```

```
Rows: 1,415
Columns: 10
$ url        <chr> "https://www.reddit.com/r/ArtificialInteligence/comments/1o~
$ author     <chr> "AutoModerator", "Baffin622", "steveb858", "PieOhMy69420", ~
$ date       <chr> "2025-10-22", "2025-10-22", "2025-10-22", "2025-10-23", "20~
$ timestamp  <dbl> 1761157899, 1761158234, 1761160658, 1761182798, 1761194796,~
$ score      <dbl> 1, 1056, 197, 129, 96, 36, 27, 3, 2, 2, 2, 17, 24, 2, 2, 9,~
$ upvotes    <dbl> 1, 1056, 197, 129, 96, 36, 27, 3, 2, 2, 2, 17, 24, 2, 2, 9,~
$ downvotes  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ golds      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ comment    <chr> "## Welcome to the r/ArtificialIntelligence gateway### Ques~
$ comment_id <chr> "1", "2", "2_1", "2_1_1", "2_1_1_1", "2_1_1_1_1", "2_1_1_1_~
```

```r
# This is all comments in the top three threads
glimpse(df_top3_comment$comment)
```

```
 chr [1:1415] "## Welcome to the r/ArtificialIntelligence gateway### Question Discussion Guid
Please use the following"| __truncated__ ...
```

a. Combine all the comments from the top three threads collected in Question 1. Pre-process the comments (remove stopwords, punctuation, etc.) and construct a Document Term Matrix using TF-IDF weighting.

```r
library("tm")
```

```
Loading required package: NLP


Attaching package: 'NLP'


The following object is masked from 'package:ggplot2':

    annotate
```

```r
library("SnowballC")
redditt.corpus = Corpus(VectorSource(df_top3_comment$comment))
redditt.corpus = tm_map(redditt.corpus,
                        function(x) iconv(x, to='UTF8', sub='byte'))
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, function(x) iconv(x, to =
"UTF8", : transformation drops documents
```

Pre-processing

```r
redditt.corpus = tm_map(redditt.corpus, function(x) iconv(x, to='ASCII', sub=' '))
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, function(x) iconv(x, to =
"ASCII", : transformation drops documents
```

```r
# remove numbers
redditt.corpus = tm_map(redditt.corpus, removeNumbers)
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, removeNumbers): transformation
drops documents
```

```r
# remove punctuation
redditt.corpus = tm_map(redditt.corpus, removePunctuation)
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, removePunctuation):
transformation drops documents
```

```r
# remove whitespace
redditt.corpus = tm_map(redditt.corpus, stripWhitespace)
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, stripWhitespace): transformation
drops documents
```

```r
# convert all to lowercase
redditt.corpus = tm_map(redditt.corpus, tolower)
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, tolower): transformation drops
documents
```

```r
# remove stopwords
redditt.corpus = tm_map(redditt.corpus, removeWords, stopwords())
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, removeWords, stopwords()):
transformation drops documents
```

```r
# convert all words to their stems
redditt.corpus = tm_map(redditt.corpus, stemDocument)
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, stemDocument): transformation
drops documents
```

```r
# remove special characters
redditt.corpus = tm_map(redditt.corpus, function(x) iconv(x, to='ASCII', sub=' '))
```

```
Warning in tm_map.SimpleCorpus(redditt.corpus, function(x) iconv(x, to =
"ASCII", : transformation drops documents
```

Creating Document Term Matrix using TF-IDF

```r
# create a term document matrix
redditt.tdm =TermDocumentMatrix(redditt.corpus)
# compute TF-IDF
redditt.wtdm = weightTfIdf(redditt.tdm)
```

```
Warning in weightTfIdf(redditt.tdm): empty document(s): 8 447 496
```

```r
# change tdm to a document term matrix
redditt.matrix = t(as.matrix(redditt.wtdm))
# remove empty tweets
empties = which(rowSums(abs(redditt.matrix)) == 0)
redditt.matrix = redditt.matrix[-empties,]
dim(redditt.matrix)
```

```
[1] 1412 5456
```

b. According to your group, what is the optimal number of clusters for these comments? Use the Elbow Method analysis to justify your argument mathematically. Explain your reasoning in detail.

Elbow Method analysis

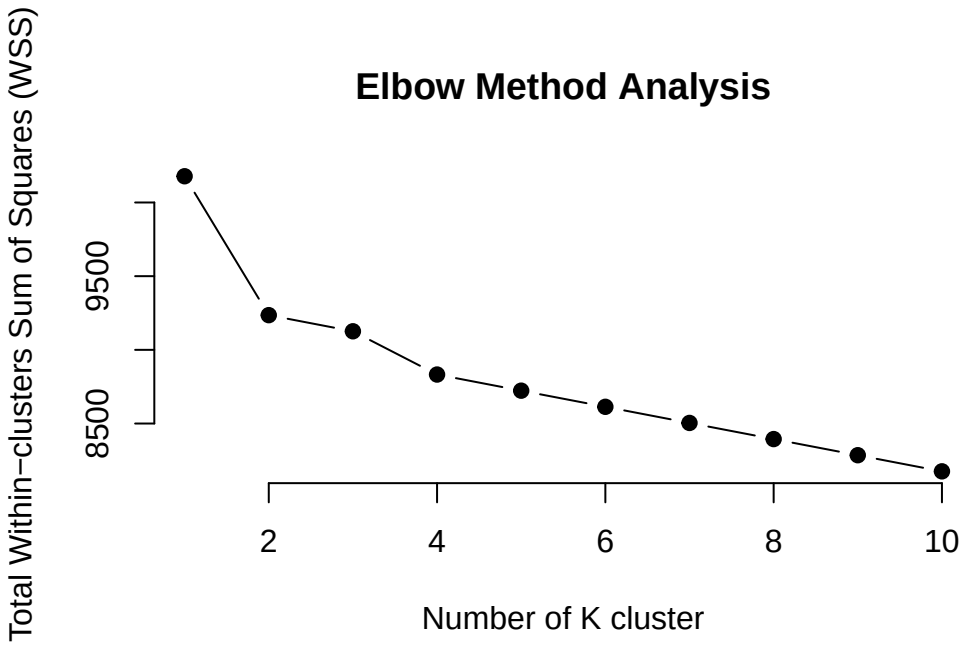
```
library(Matrix)
```

```
Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
reddit.matrix.sparse <- as(redditt.matrix, "sparseMatrix")
```

```
# Convert to dense
reddit_tfidf_matrix <- as.matrix(reddit.matrix.sparse)

# Define a random k number
k_max <- 10
# Caculate Total Within Sum of Square
wss <- sapply(1:k_max, function(k){
  kmeans(reddit_tfidf_matrix, centers = k, nstart = 5, iter.max = 50)$tot.withinss
})
```

```
# Plotting the graph
plot(1:k_max, wss,
     type = "b", pch = 19, frame = FALSE,
     xlab = "Number of K cluster",
     ylab = "Total Within-clusters Sum of Squares (WSS)",
     main = "Elbow Method Analysis")
```



Based on the Elbow Method Analysis graph, it is observed that the Total Within Cluster Sum of Squares (WSS) value decreases most sharply as the number of clusters increases from $k = 1$ to $k = 3$. From $k = 3$ onwards, although WSS continues to decrease, the rate of decrease slows down significantly, forming a distinct 'elbow point' at $k = 2$. Therefore, we determine the optimal number of clusters for classifying Reddit comments in this problem to be 3.

c. Create a "Ground Truth" label for each comment based on its original thread.

- Label Thread 1 comments as "A"
- Label Thread 2 comments as "B"
- Label Thread 3 comments as "C"

```
library(dplyr)
df_top3_comment <- df_top3_comment %>%
  mutate(ground_truth = case_when(
    url == "https://www.reddit.com/r/ArtificialInteligence/comments/1odgfys/i_was_once_an_ai_
    url == "https://www.reddit.com/r/ArtificialInteligence/comments/1orewim/meta_just_lost_20
    url == "https://www.reddit.com/r/ArtificialInteligence/comments/1kb8e09/i_lost_my_busines
    TRUE ~ "Unknown"
  ))
table(df_top3_comment$ground_truth) #Check again
```

```
  A   B   C
467 462 486
```

d. Apply K-means clustering to the combined dataset. You may need to use a dimension reduction method (PCA or MDS) before applying the K-means method to improve performance.

```r
# Dimension reduction (PCA)
thread_pca <- prcomp(reddit_tfidf_matrix)
pca_data <- as.data.frame(thread_pca$x[, 1:4]) # Check first 4 components
head(pca_data, 30)
```

```
          PC1         PC2           PC3           PC4
1  -0.1311449 -0.03018987 -1.741775e-15 -4.435142e-17
2  -0.1319634 -0.03038083 -1.791215e-15 -3.808878e-16
3  -0.1313643 -0.03038541 -2.263927e-15  4.023399e-16
4  -0.1318057 -0.03067718 -1.810297e-15  5.019101e-17
5  -0.1330268 -0.03165827 -2.704547e-15  4.162177e-16
6  -0.1314314 -0.03038187 -1.132020e-15 -6.905359e-16
7  -0.1326243 -0.03127389 -1.837185e-15 -2.152217e-16
9  -0.1315780 -0.03050424 -1.583048e-15 -8.251534e-17
10 -0.1307721 -0.02986753 -1.533608e-15 -4.782087e-17
11 -0.1361390 -0.03410295 -2.485104e-15 -2.873891e-17
12 -0.1320550 -0.03093653 -1.988973e-15  1.881015e-16
13 -0.1333467 -0.03183587 -1.613406e-15  9.270937e-16
14 -0.1300120 -0.02927500 -1.602997e-15  2.791745e-16
15 -0.1330684 -0.03164247 -2.060964e-15  7.690280e-18
16 -0.1318287 -0.03065737 -1.970759e-15  3.364204e-16
17 -0.1330890 -0.03166162 -1.948207e-15  8.575284e-17
18 -0.1331634 -0.03169129 -2.087852e-15 -2.351710e-16
19 -0.1306340 -0.02978095 -1.802490e-15  6.840560e-17
20 -0.1308953 -0.02997204 -1.620345e-15 -3.670100e-16
21 -0.1342994 -0.03240478 -2.061832e-15  8.748756e-17
22 -0.1332840 -0.03177990 -1.848461e-15 -3.307572e-17
23 -0.1305103 -0.02971035 -1.790347e-15  3.353472e-18
24 -0.1339263 -0.03229843 -2.046219e-15 -2.082828e-16
25 -0.1320700 -0.03089647 -1.895298e-15  1.013653e-16
26 -0.1333264 -0.03189403 -2.225763e-15  1.534071e-16
27 -0.1356150 -0.03271222 -2.414848e-15  1.239168e-16
28 -0.1304520 -0.02950314 -1.903972e-15 -1.050667e-16
```

```
29 -0.1309860 -0.03006452 -1.705346e-15  7.534450e-17
30 -0.1314652 -0.03043540 -1.773000e-15  8.141603e-17
31 -0.1321680 -0.03090257 -1.879686e-15  7.621186e-17
```

As seen in the PCA results, the components have extremely small values, which ranging from 10^{-15} to 10^{-17}, suggesting that the variance cannot be fully captured by using the PCA since the data is scattered to show clear separation. Hence, it is vital to use the multidimensional scaling (MDS) for text data for it will preserve the distance between comments.

```r
# Dimension reduction (MDS)
library(proxy)
```

```
Attaching package: 'proxy'

The following object is masked from 'package:Matrix':

    as.matrix

The following objects are masked from 'package:stats':

    as.dist, dist

The following object is masked from 'package:base':

    as.matrix
```

```r
# Calculate Cosine Distance Matrix
distance_matrix <- proxy::dist(reddit_tfidf_matrix, method = "cosine")
```

```r
# Run MDS on cmdscale
# With k = 2, reduce the dataset to 2 dimensions
thread_mds <- cmdscale(distance_matrix, k = 2)

# Create a data frame for clustering and plotting
threat_mds_data <- as.data.frame(thread_mds)
colnames(threat_mds_data) <- c("Dim1", "Dim2")
```

```r
# Apply K-means clustering with k=3
set.seed(123)
k.mean.value <- 3
thread_clusters <- kmeans(threat_mds_data, centers = k.mean.value, nstart = 25)
```

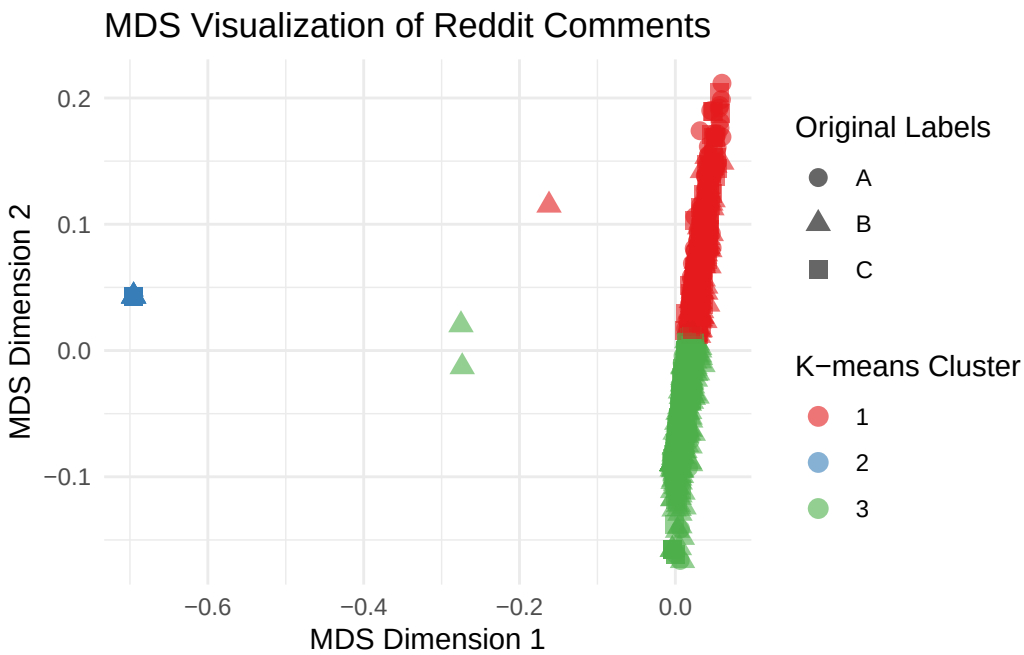
```r
names(thread_clusters)
```

```
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

```r
# Combine the result with ground truth labels
threat_mds_data$cluster <- as.factor(thread_clusters$cluster)


num_rows <- as.numeric(rownames(threat_mds_data)) # Identify available rows in MDS data
matched_value <- df_top3_comment$ground_truth[num_rows] # Apply labels with those rows
threat_mds_data$ground_truth <- factor(matched_value, labels = c("A", "B", "C"))
```

e. Visualize the results of your clustering in a 2D vector space. Color the points by their assigned cluster but use different markers (shapes) to represent their original thread label.

```r
library(ggplot2)

ggplot(threat_mds_data, aes(x = Dim1, y = Dim2, color = cluster, shape = ground_truth)) +
  geom_point(size = 3, alpha = 0.6) +
  theme_minimal() +
  labs(title = "MDS Visualization of Reddit Comments",
       x = "MDS Dimension 1",
       y = "MDS Dimension 2",
       color = "K-means Cluster",
       shape = "Original Labels") +
  scale_color_brewer(palette = "Set1")
```



f. Comment on your findings. Did the K-means algorithm successfully reconstruct the original thread boundaries? Discuss any overlaps observed in the visualization.

- **Thread boundaries:** The K-means is not suitable for reconstructing the thread boundaries. The threads labels with different shapes are scattered among the three groups, and are not visibly distinguished within the clusters.
- **Visualization overlaps:** Along the right side of the plot, there is a significant amount of vertical overlap.
  - The dense vertical line 0.0 indicates that the majority of comments in all three threads use a highly similar vocabulary. With the word artificial intelligence, this probably indicates that users were using the same terms, such as AI, model, or data, across all threads, making it challenging for the algorithm to identify a clear mathematical difference across the discussions.
  - Cluster red represents a small, distinct comments that are separated. These are distinct comments that differ greatly from the rest of the dataset in terms of vocabulary, for example, technical discussions, short comments, or irrelevant spam.
  - Rather of dividing the AI conversation by their original thread labels, the clusters blue and green effectively divided it in half based on little differences in attitude or subtopics.

"Artificial Intelligence" seems to be a frequent subject in these Reddit threads. This overlap implies that the discussions within the posts are linguistically similar using the K-means model, even though their titles may differ. Since K-means is a central-based approach, comments in Threads A, B, and C overlap in the MDS graph because the mathematical distance between them is almost zero.

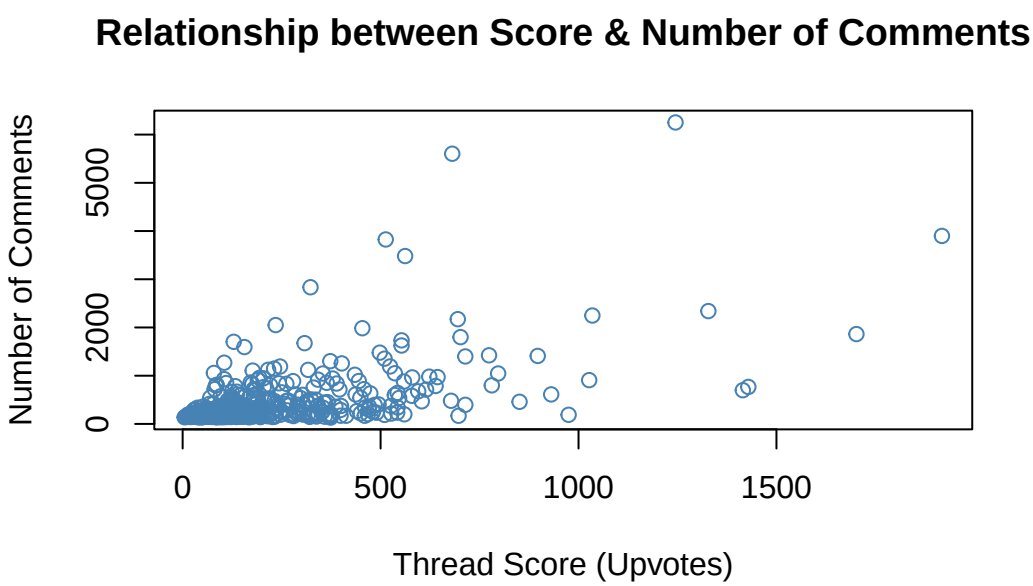
Question 3

```
dim(df_only_threads)
```

```
[1] 437  15
```

Scatter plot

```
plot(df_only_threads$comments, df_only_threads$score,
     main = "Relationship between Score & Number of Comments",
     xlab = "Thread Score (Upvotes)",
     ylab = "Number of Comments",
     pch = 1,
     col = "steelblue")
```



Comment on scatter plot:

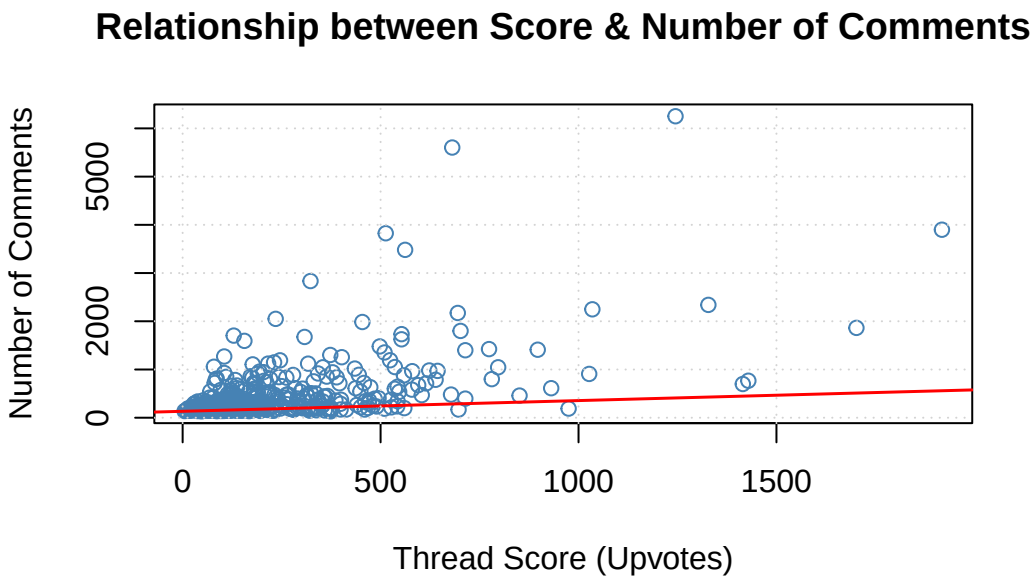
According to the scatter plot, it can be seen that there is a positive linear relationship between thread scores and the number of comments. This means that the higher a post's score, the greater the likelihood of it receiving more attention and discussion from the community.

However, the data points are highly dispersed, especially in the group of posts with high scores. Notably, there are outliers that show extremely high engagement. For example, a thread with a score of only around 1,200 attracted over 6,000 comments (the highest in the dataset). Conversely, some threads with very high scores (above 1,500) had significantly lower discussion volumes compared to posts with average scores.

Although the scatter plot shows a positive relationship visually, to validate this relationship and to measure the specific impact of scores on comment volume, Linear Regression will be implemented in the following section.

Linear Regression

```r
# Run a linear regression model.
lm_reg <- lm(comments ~ score, data = df_only_threads)
# Plot the linear regression line
plot(df_only_threads$comments, df_only_threads$score,
     main = "Relationship between Score & Number of Comments",
     xlab = "Thread Score (Upvotes)",
     ylab = "Number of Comments",
     pch = 1,
     col = "steelblue")
abline(lm_reg, col = "red", lwd = 1.5)
grid(col = "lightgray", lty = "dotted")
```




```r
summary(lm_reg)
```

```
Call:
lm(formula = comments ~ score, data = df_only_threads)

Residuals:
    Min      1Q  Median      3Q     Max 
-697.15 -115.90  -55.12   61.88 1154.90 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 133.12193   12.06832   11.03   <2e-16 ***
score         0.22221    0.01589   13.98   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 200.1 on 435 degrees of freedom
Multiple R-squared:  0.3101,	Adjusted R-squared:  0.3085 
F-statistic: 195.5 on 1 and 435 DF,  p-value: < 2.2e-16
```

Comment on the regression result

Overall, the regression model achieved a statistical value of F = 195.5 with a p-value < 2.2e-16. Since this value is below the significance level of 0.05, it can conclude that the model is

statistically significant and the Score variable is an important predictor of the number of comments.

Specifically, the regression results show that the score variable has a positive and significant impact on the number of comments (since $p < 2e\text{-}16$). Particularly, the regression coefficient of the score variable is 0.22221, which means that, all other factors being constant, when a thread gains 1 Upvote point, it is expected to receive approximately 0.22 more comments. The extremely small p-value of the score variable indicates that this relationship is not random.

Moreover, regarding the model's explanatory ability, the Multiple R-squared value reached 0.3101. This indicates that approximately 31% of the variation in the number of comments can be explained by the post's score. Similarly, the Adjusted R-squared value reached a comparable level of 0.3085. The close difference between these two indicators suggests that the current simple regression model is quite stable and not overfitting.

In conclusion, the linear regression results confirm that thread scores have a positive influence on the number of comments. However, since the R-square is quite low (31%), it can be seen that the score is not the only factor determining the level of interaction. Therefore, a post with a high Upvotes score does not necessarily guarantee a corresponding amount of discussion, as engagement on Reddit is also strongly influenced by other factors such as content and topic characteristics.

Question 4:

a. Using the Mastodon API, identify users related to your chosen topic. Identify the top three most popular users based on the highest number of Statuses (Posts) (i.e., the most active users, rather than just the most followed).

```r
library(purrr)
library(dplyr)
library(tidyverse)
library(rtoot)
```

```
Warning: package 'rtoot' was built under R version 4.4.3
```

```r
# Connect with the Mastodon account
auth_setup(instance = "mastodon.social", type = "user")

# Search accounts related to the topic of AI
mastodon_ai_accounts <- search_accounts("Artificial Intelligence")
```

```r
# Save to the file for further use
saveRDS(mastodon_ai_accounts, "mastodon_ai_accounts.rds")
```

```r
mastodon_ai_accounts <- readRDS("mastodon_ai_accounts.rds")
dim(mastodon_ai_accounts)
```

```
[1] 40 21
```

There are 40 accounts related to this topic of "Artificial Intelligence".

```r
# Look at the note to see what contents about the AI topic
mastodon_ai_accounts$note[1:10]
```

```
 [1] "<p>  Artificial Intelligence News<br>  Artificial Intelligence Updates<br>  Artificial I
 [2] "<p> AI Expert | Tech Innovations<br> Unlocking AI's Potential: ML, Robotics &amp; More<
 [3] "<p>  Designer  Art Director<br> Worldbuilder   Researcher</p>"
 [4] "<p>Artificial Intelligence News, Art and Music.<br />I am John Miller, @aibrief and <sp
card\" translate=\"no\"><a href=\"https://mastodon.social/@millerfilm\" class=\"u-
url mention\">@<span>millerfilm</span></a></span> on X/Twitter.<br /><a href=\"https://masto
 [5] "<p>Generative artist, poet, &amp; developer<br>sharing \"Bittersweet Hearts\" in parts
 [6] "<p>Machine learning is transforming our lives in nearly every way. Here are the most in
intelligence-fp5efm1rz\" rel=\"nofollow noopener\" target=\"_blank\">Artificial Intelligence
card\" translate=\"no\"><a href=\"https://flipboard.com/@WIRED\" class=\"u-
url mention\" rel=\"nofollow noopener\" target=\"_blank\">@<span>WIRED</span></a></span>.</p
 [7] "<p>Espionage, Spycraft, Spies, Whistleblowers and leakers</p>\n"
 [8] "<p>Generative artificial intelligence tools such as ChatGPT and Sora can be used to pro
intelligence-ai-literacy-and-ai-falsehoods-tb21av82z\" rel=\"nofollow noopener\" target=\"_bl
card\" translate=\"no\"><a href=\"https://flipboard.com/@NewsLitProject\" class=\"u-
url mention\" rel=\"nofollow noopener\" target=\"_blank\">@<span>NewsLitProject</span></a></
 [9] "<p>For too long the hoarders of resources and inept systems have dictated that ordinary
sustaining actions. It is time for a global political and economical revolution. For a truly
[10] "<p>The hottest topic in tech: Here's everything you need to know about ChatGPT, Bard, D
E and more.</p><p><a href=\"https://flipboard.com/@gizmodo/artificial-intelligence-
7bv07f1uz\" rel=\"nofollow noopener\" target=\"_blank\">Artificial intelligence</a> is a Flip
card\" translate=\"no\"><a href=\"https://flipboard.com/@Gizmodo\" class=\"u-
url mention\" rel=\"nofollow noopener\" target=\"_blank\">@<span>Gizmodo</span></a></span>.<
```

As can be seen, the topic of Artificial Intelligence contains not only the discussion on technology, but also about the usage of AI in different areas such as art, machine learning, along with AI news.

```r
names(mastodon_ai_accounts)
```

```
 [1] "id"               "username"         "acct"             "display_name"
 [5] "locked"           "bot"              "discoverable"     "group"
 [9] "created_at"       "note"             "url"              "avatar"
[13] "avatar_static"    "header"           "header_static"    "followers_count"
[17] "following_count" "statuses_count"  "last_status_at"  "fields"
[21] "emojis"
```

```r
# Top 3 most popular users
top_3_users <- mastodon_ai_accounts %>%
  distinct(mastodon_ai_accounts$id, .keep_all = TRUE) %>%
  arrange(desc(mastodon_ai_accounts$statuses_count)) %>%
  slice(1:3)

# View the IDs of your top 3
print(top_3_users)
```

```
# A tibble: 3 x 22
  id                 username acct  display_name locked bot   discoverable group
  <chr>              <chr>    <chr> <chr>        <lgl>  <lgl> <lgl>        <lgl>
1 112979380340201604 artific~ arti~ Artificial ~ FALSE  FALSE FALSE        FALSE
2 112260083730143404 artific~ arti~ Artificial ~ FALSE  FALSE TRUE         TRUE
3 1099674            BrynSim  Bryn~ artificial ~ FALSE  TRUE  FALSE        FALSE
# i 14 more variables: created_at <dttm>, note <chr>, url <chr>, avatar <chr>,
#   avatar_static <chr>, header <chr>, header_static <chr>,
#   followers_count <int>, following_count <int>, statuses_count <int>,
#   last_status_at <dttm>, fields <I<list>>, emojis <I<list>>,
#   `mastodon_ai_accounts$id` <chr>
```

```r
top_3_users$statuses_count
```

```
[1] 11125 10782  9059
```

```r
top_3_users$followers_count
```

```
[1] 886  13  27
```

```r
top_3_users$following_count
```

```
[1] 191   0   2
```

These top 3 accounts have the highest number of status counts, which 11031, 10782, and 9059 respectively. However, considering the number of followers and the following count, there are 2 accounts which have no or limited followers and following number, such as the account "artificial-intelligence-machine-learning-MattMagnuson@flipboard.com" and "BrynSim@botsin.space". This is because these accounts may be automation account using bot ("BrynSim"), privacy settings, or new accounts which imported posts from other social platforms but has not build the network on Mastodon. Hence, it is important to filter these accounts.

```r
# Define the minimum number of followers and following
minimum_followers <- 50
minimum_following <- 50
```

```r
# Top 3 most popular users
top_3_active_users <- mastodon_ai_accounts %>%
  filter(followers_count >= minimum_followers,
         following_count >= minimum_following) %>%
  distinct(id, .keep_all = TRUE) %>%
  arrange(desc(statuses_count)) %>%
  slice(1:3)

# View the IDs of your top 3
print(top_3_active_users)
```

```
# A tibble: 3 x 21
  id                 username acct  display_name locked bot   discoverable group
  <chr>              <chr>    <chr> <chr>        <lgl>  <lgl> <lgl>        <lgl>
1 112979380340201604 artific~ arti~ Artificial ~ FALSE  FALSE FALSE        FALSE
2 116097066760715533 fernand~ fern~ Inteligênci~ FALSE  FALSE FALSE        FALSE
3 109626341387797094 j_hears~ j_he~ Articulated~ FALSE  FALSE FALSE        FALSE
# i 13 more variables: created_at <dttm>, note <chr>, url <chr>, avatar <chr>,
#   avatar_static <chr>, header <chr>, header_static <chr>,
#   followers_count <int>, following_count <int>, statuses_count <int>,
#   last_status_at <dttm>, fields <I<list>>, emojis <I<list>>
```

```r
top_3_active_users$acct
```

```
[1] "artificialsweetener.ai@threads.net" "fernando.gpt@threads.net"
[3] "j_hearsay@infosec.exchange"
```

Following this result, the first two accounts have the ending with 'threads.net', and these accounts will derive followers and followings from another social media platforms, and this will show null value when analyze network on Mastodon. Thus, it is vital to remove this type of account for better network analysis result.

```r
# Top 3 most popular users
top_3_active_users <- mastodon_ai_accounts %>%
  filter(!str_detect(acct, "threads.net")) %>%
  # Avoid threads accounts which derived followers and friends from other social platforms
  filter(followers_count >= minimum_followers,
         following_count >= minimum_following) %>%
  distinct(id, .keep_all = TRUE) %>%
  arrange(desc(statuses_count)) %>%
  slice(1:3)

# View the IDs of your top 3
print(top_3_active_users)
```

```
# A tibble: 3 x 21
  id                 username acct  display_name locked bot   discoverable group
  <chr>              <chr>    <chr> <chr>        <lgl>  <lgl> <lgl>        <lgl>
1 109626341387797094 j_hears~ j_he~ Articulated~ FALSE  FALSE FALSE        FALSE
2 748749             artific~ arti~ Rafa Gálvez  FALSE  FALSE TRUE         FALSE
3 109291448269519682 artific~ arti~ Andy Goralc~ FALSE  FALSE TRUE         FALSE
# i 13 more variables: created_at <dttm>, note <chr>, url <chr>, avatar <chr>,
#   avatar_static <chr>, header <chr>, header_static <chr>,
#   followers_count <int>, following_count <int>, statuses_count <int>,
#   last_status_at <dttm>, fields <I<list>>, emojis <I<list>>
```

```r
top_3_active_users$acct
```

```
[1] "j_hearsay@infosec.exchange"
[2] "artificialphilosopher@scholar.social"
[3] "artificial3d@mastodon.art"
```

```r
top_3_active_users$statuses_count
```

```
[1] 4203 2563  185
```

```r
top_3_active_users$followers_count
```

```
[1] 222 304 887
```

```r
top_3_active_users$following_count
```

```
[1] 1856  121  162
```

```r
top_3_active_users$acct
```

```
[1] "j_hearsay@infosec.exchange"
[2] "artificialphilosopher@scholar.social"
[3] "artificial3d@mastodon.art"
```

```r
top_3_active_users$username
```

```
[1] "j_hearsay"             "artificialphilosopher" "artificial3d"
```

```r
top_3_active_users$note
```

```
[1] "<p>If there's been a way to build it, there'll be a way to destroy it. Things are not al
[2] "<p>I am a privacy engineer passionate about artificial intelligence and its potential to
card\" translate=\"no\"><a href=\"https://mastodon.social/@CosicBe\" class=\"u-
url mention\" rel=\"nofollow noopener\" target=\"_blank\">@<span>CosicBe</span></a></span>, I
[3] "<p>Art Director at <a href=\"https://mastodon.art/tags/blender\" class=\"mention hashtag
```

As can be seen, these users are not strictly focused on AI but a combination of different usages of AI in many fields and technology.

- The first user is a personal account and giving a philosophical aspect, an information explorer, towards the combination of technology, creativity, and critical thinking. With a very high number of 4203 posts, high friends (1856) and followers (222), this account follows a wide users to gather diverse information, and also connect with various communities. A lower follower counts shows that the influence of this account is highly connected.

- The second account is an engineer with a professional specialty in artificial intelligence and a strong academic background. This user has a PhD about AI engineering, and currently focus on balancing AI innovation with privacy protections, and giving high-impact research in real world situation. The second user has more followers than friends, 304 and 121 respectively, indicating that they target a high quality audience.
- The third user is an experienced art director for movie projects, and may be an influencer in the creative community. This individual has several contributions to various animated films, and has a proficient background in creative leadership, animation, and digital art. This account has lower posts of 185 but has highest followers of 887, suggested high value contents of creative AI visuals in digital art.

b. Download 50 followers and 50 friends (users they follow) for these top 3 active users. Create a graph and visualize the network structure.

```
# Create a list to store the result
network_list <- list()

# Using loops to extract 50 followers and 50 friends of the top accounts
for (i in 1:nrow(top_3_active_users)) {
  # Extract current user details
  account_id <- top_3_active_users$id[i]
  account_name <- top_3_active_users$username[i]

  # Get 50 followers
  followers <- get_account_followers(account_id, limit = 50)

    # Check if data exists and has columns
    if (length(followers) > 0 && "acct" %in% names(followers)) {
      topacc_followers <- followers %>%
        mutate(type = "follower",
               source = acct,
               target = account_name) %>%
        select(source, target, type)
    } else {
      topacc_followers <- data.frame(source=character(),
                                     target=character(),
                                     type=character())
    }
  # Get 50 friends
    friends <- get_account_following(account_id, limit = 50)
```

```r
    if (length(friends) > 0 && "acct" %in% names(friends)) {
      topacc_friends <- friends %>%
        mutate(type = "friend",
               source = account_name,
               target = acct) %>%
        select(source, target, type)
    } else {
      topacc_friends <- data.frame(source=character(),
                                   target=character(),
                                   type=character())
    }

    # Combine the results for each user
    network_list[[i]] <- bind_rows(topacc_followers, topacc_friends)
    Sys.sleep(1) # Shorten loading time
}

saveRDS(network_list, "network_data.rds")
```

```r
network_data <- readRDS("network_data.rds")
head(network_data)
```

```
[[1]]
# A tibble: 134 x 3
   source              target    type
   <chr>               <chr>     <chr>
 1 Silvia_Project      j_hearsay follower
 2 novafuture          j_hearsay follower
 3 nada_eid_88         j_hearsay follower
 4 Hodhod_233          j_hearsay follower
 5 bilalmohammedzeyada j_hearsay follower
 6 knowprose           j_hearsay follower
 7 save_tocat199       j_hearsay follower
 8 fllics              j_hearsay follower
 9 almiramat           j_hearsay follower
10 1div0               j_hearsay follower
# i 124 more rows

[[2]]
# A tibble: 88 x 3
   source                target                type
```

```
   <chr>                  <chr>                  <chr>
 1 amazing_Subzy5588      artificialphilosopher follower
 2 str4t0tt0              artificialphilosopher follower
 3 rezaebrahimiatani      artificialphilosopher follower
 4 dongcarl               artificialphilosopher follower
 5 cyberthreatdefence     artificialphilosopher follower
 6 jessicabennet          artificialphilosopher follower
 7 angeliquetrusler       artificialphilosopher follower
 8 thierry_vandenbussche artificialphilosopher follower
 9 tankz13                artificialphilosopher follower
10 baglunch               artificialphilosopher follower
# i 78 more rows

[[3]]
# A tibble: 116 x 3
   source           target       type
   <chr>            <chr>        <chr>
 1 hbons            artificial3d follower
 2 tjdiaz           artificial3d follower
 3 aihacks          artificial3d follower
 4 element_112      artificial3d follower
 5 3D_efluencer     artificial3d follower
 6 ruvaldese        artificial3d follower
 7 ruthkalypsoart   artificial3d follower
 8 alejandrosoriano artificial3d follower
 9 calver           artificial3d follower
10 sreeragpraful    artificial3d follower
# i 106 more rows
```

```r
# Combine the results to an edge list
all_acc_edges <- bind_rows(network_data)

# View the data
print(all_acc_edges, n=150)
```

```
# A tibble: 338 x 3
    source                target                type
    <chr>                 <chr>                 <chr>
  1 Silvia_Project        j_hearsay             follower
  2 novafuture            j_hearsay             follower
  3 nada_eid_88           j_hearsay             follower
  4 Hodhod_233            j_hearsay             follower
```

5	bilalmohammedzeyada	j_hearsay	follower
6	knowprose	j_hearsay	follower
7	save_tocat199	j_hearsay	follower
8	fllics	j_hearsay	follower
9	almiramat	j_hearsay	follower
10	1div0	j_hearsay	follower
11	shaharyarranjah1	j_hearsay	follower
12	BrewtBlacklist	j_hearsay	follower
13	garywetzel	j_hearsay	follower
14	rodaina85	j_hearsay	follower
15	lina19942810	j_hearsay	follower
16	hebaakram	j_hearsay	follower
17	israajamal	j_hearsay	follower
18	mohammadizadeh	j_hearsay	follower
19	ahmedaziz24852	j_hearsay	follower
20	wafaagaza1499	j_hearsay	follower
21	ameengaza	j_hearsay	follower
22	hassangaza299	j_hearsay	follower
23	emanahmed	j_hearsay	follower
24	maram4	j_hearsay	follower
25	nour_s	j_hearsay	follower
26	emma4	j_hearsay	follower
27	doaagaza	j_hearsay	follower
28	nourgaza	j_hearsay	follower
29	soma23	j_hearsay	follower
30	Felipe192	j_hearsay	follower
31	joyousjoyness	j_hearsay	follower
32	versiqcontent	j_hearsay	follower
33	nikkilafae12	j_hearsay	follower
34	CelloMomOnCars	j_hearsay	follower
35	vhtc	j_hearsay	follower
36	mattiacecchi	j_hearsay	follower
37	InamAfridi	j_hearsay	follower
38	RealJournalism	j_hearsay	follower
39	eribosot	j_hearsay	follower
40	democratsabroad	j_hearsay	follower
41	dawah_mission	j_hearsay	follower
42	DrALJONES	j_hearsay	follower
43	gogoarapantoglou	j_hearsay	follower
44	clintruin	j_hearsay	follower
45	Barthez	j_hearsay	follower
46	tom_andraszek	j_hearsay	follower
47	camelliakyoto	j_hearsay	follower

48	kcarruthers	j_hearsay	follower
49	aem76us	j_hearsay	follower
50	ftuttfy9	j_hearsay	follower
51	jcolag	j_hearsay	follower
52	BionicCity_Verse	j_hearsay	follower
53	jack_daniel	j_hearsay	follower
54	fak3r	j_hearsay	follower
55	j_hearsay	mathiasverraes	friend
56	j_hearsay	romeu	friend
57	j_hearsay	Cyriux	friend
58	j_hearsay	jcreed	friend
59	j_hearsay	JoBlakely	friend
60	j_hearsay	PattyHanson	friend
61	j_hearsay	AltEvil	friend
62	j_hearsay	QasimRashid	friend
63	j_hearsay	wearenew_public	friend
64	j_hearsay	treyhunner	friend
65	j_hearsay	peternlewis	friend
66	j_hearsay	mntmn	friend
67	j_hearsay	novafuture	friend
68	j_hearsay	nf3xn	friend
69	j_hearsay	luckytran	friend
70	j_hearsay	rgc	friend
71	j_hearsay	oskargroth	friend
72	j_hearsay	lambdageek	friend
73	j_hearsay	ronbronson	friend
74	j_hearsay	sturmsucht	friend
75	j_hearsay	Nfinitywaltz	friend
76	j_hearsay	boodoo	friend
77	j_hearsay	CatherineBabault	friend
78	j_hearsay	PythonPeak	friend
79	j_hearsay	knowprose	friend
80	j_hearsay	ScienceCommunicator	friend
81	j_hearsay	dabeaz	friend
82	j_hearsay	iFixit	friend
83	j_hearsay	ashleygjovik	friend
84	j_hearsay	randahl	friend
85	j_hearsay	gnomon	friend
86	j_hearsay	tommorris	friend
87	j_hearsay	JamesWidman	friend
88	j_hearsay	1div0	friend
89	j_hearsay	snap2grid	friend
90	j_hearsay	publicdomainrev	friend

91	j_hearsay	r_alb	friend
92	j_hearsay	daringfireball	friend
93	j_hearsay	pteranodo	friend
94	j_hearsay	dashremover	friend
95	j_hearsay	designspells	friend
96	j_hearsay	claushoumann	friend
97	j_hearsay	damieng	friend
98	j_hearsay	Ange	friend
99	j_hearsay	lukhash	friend
100	j_hearsay	OpenMediaOrg	friend
101	j_hearsay	cybertipsguide	friend
102	j_hearsay	brouhaha	friend
103	j_hearsay	BrewtBlacklist	friend
104	j_hearsay	novaTopFlex	friend
105	j_hearsay	3psboyd	friend
106	j_hearsay	murena	friend
107	j_hearsay	mcnees	friend
108	j_hearsay	raiderrobert	friend
109	j_hearsay	passbolt	friend
110	j_hearsay	tveskov	friend
111	j_hearsay	WrtrStat	friend
112	j_hearsay	ctietze	friend
113	j_hearsay	tbernard	friend
114	j_hearsay	ekis	friend
115	j_hearsay	brettcannon	friend
116	j_hearsay	privacytests	friend
117	j_hearsay	etchedpixels	friend
118	j_hearsay	superbetsy	friend
119	j_hearsay	mistakenotmy	friend
120	j_hearsay	carlysagan	friend
121	j_hearsay	kangaroo5383	friend
122	j_hearsay	_elena	friend
123	j_hearsay	andrewrk	friend
124	j_hearsay	Rachelburch	friend
125	j_hearsay	cmconseils	friend
126	j_hearsay	grwster	friend
127	j_hearsay	jensorensen	friend
128	j_hearsay	doriantaylor	friend
129	j_hearsay	ethanschoonover	friend
130	j_hearsay	richpuchalsky	friend
131	j_hearsay	joyousjoyness	friend
132	j_hearsay	ASegar	friend
133	j_hearsay	runbox	friend

```
134 j_hearsay             JasperSiepkes         friend
135 amazing_Subzy5588     artificialphilosopher follower
136 str4t0tt0             artificialphilosopher follower
137 rezaebrahimiatani     artificialphilosopher follower
138 dongcarl              artificialphilosopher follower
139 cyberthreatdefence    artificialphilosopher follower
140 jessicabennet         artificialphilosopher follower
141 angeliquetrusler      artificialphilosopher follower
142 thierry_vandenbussche artificialphilosopher follower
143 tankz13               artificialphilosopher follower
144 baglunch              artificialphilosopher follower
145 oneguynick            artificialphilosopher follower
146 alshlyapin            artificialphilosopher follower
147 FinnleyGraves         artificialphilosopher follower
148 nicholas_saunders     artificialphilosopher follower
149 CosicBe               artificialphilosopher follower
150 aidanonlearning       artificialphilosopher follower
# i 188 more rows
```

```r
# Simplify for the graph (Source -> Target)
acc_edges <- all_acc_edges %>%
  select(source, target)

# View the structure
print(head(acc_edges))
```

```
# A tibble: 6 x 2
  source              target
  <chr>               <chr>
1 Silvia_Project      j_hearsay
2 novafuture          j_hearsay
3 nada_eid_88         j_hearsay
4 Hodhod_233          j_hearsay
5 bilalmohammedzeyada j_hearsay
6 knowprose           j_hearsay
```

```r
library(igraph)
```

```
Attaching package: 'igraph'
```

```
The following objects are masked from 'package:lubridate':

    %--%, union

The following objects are masked from 'package:dplyr':

    as_data_frame, groups, union

The following objects are masked from 'package:purrr':

    compose, simplify

The following object is masked from 'package:tidyr':

    crossing

The following object is masked from 'package:tibble':

    as_data_frame

The following objects are masked from 'package:stats':

    decompose, spectrum

The following object is masked from 'package:base':

    union
```

```r
library(ggraph)

# Create a directed graph from the edge list
network_graph <- graph_from_data_frame(d = acc_edges, directed = TRUE)

# Print summary to verify nodes and edges
print(network_graph)
```

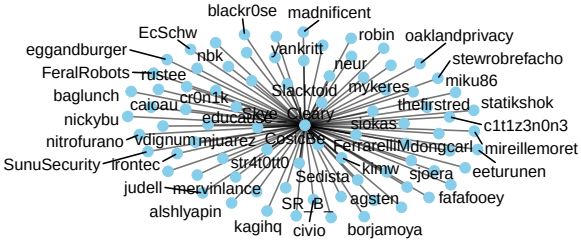
```
IGRAPH 08be464 DN-- 317 338 --
+ attr: name (v/c)
+ edges from 08be464 (vertex names):
 [1] Silvia_Project     ->j_hearsay novafuture         ->j_hearsay
```

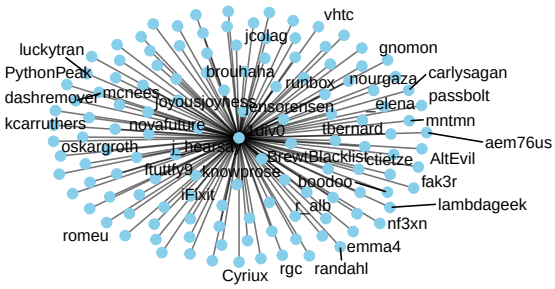
```
 [3] nada_eid_88        ->j_hearsay Hodhod_233         ->j_hearsay
 [5] bilalmohammedzeyada->j_hearsay knowprose          ->j_hearsay
 [7] save_tocat199      ->j_hearsay fllics             ->j_hearsay
 [9] almiramat          ->j_hearsay 1div0              ->j_hearsay
[11] shaharyarranjah1   ->j_hearsay BrewtBlacklist     ->j_hearsay
[13] garywetzel         ->j_hearsay rodaina85          ->j_hearsay
[15] lina19942810       ->j_hearsay hebaakram          ->j_hearsay
+ ... omitted several edges
```

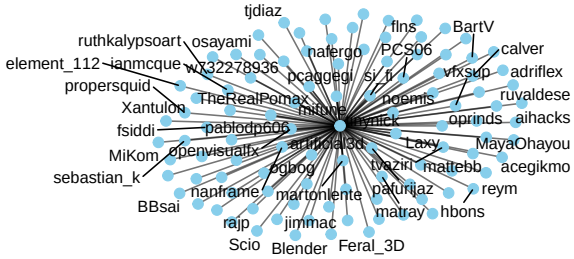
```r
ggraph(network_graph, layout = "fr") +
  # The connections
  geom_edge_link(aes(alpha = 0.3), show.legend = FALSE) +
  # The users as points
  geom_node_point(color = "skyblue", size = 3) +
  # Add labels to "Top 3" users
  geom_node_text(aes(label = name),
                 repel = TRUE,
                 size = 4,
                 max.overlaps = 10) +
  theme_graph(base_family = "sans") +
  labs(title = "Top 3 Mastodon Active Accounts Network About AI Topic")
```

```
Warning: ggrepel: 182 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

Top 3 Mastodon Active Accounts Network About AI Topic







The graph shows clear group pattern for each top user, with many surrounding nodes and connect to the central user. This indicates that each person has separated network and their own community. Furthermore, there are low interconnection between followers, showing an interested-based with a central user, loosely clusters, and not tight communities.

c. Calculate both the Degree Centrality and the PageRank for the nodes in your graph.

- Identify the top user for each metric.
- Comment on the differences: Does the user with the most connections (Degree) also have the highest influence/quality connections (PageRank)?

```
# Calculate Degree Centrality
central_degree <- degree(network_graph, mode = "all")
head(central_degree)
```

```
     Silvia_Project          novafuture          nada_eid_88          Hodhod_233
                  1                   2                    1                   1
bilalmohammedzeyada           knowprose
                  1                   2
```

```r
# Calculate Pagerank
pagerk <- page_rank(network_graph)$vector
head(pagerk)
```

```
     Silvia_Project          novafuture         nada_eid_88          Hodhod_233
        0.001415612         0.002154390         0.001415612         0.001415612
bilalmohammedzeyada           knowprose
        0.001415612         0.002154390
```

```r
# Combine Degree Centrality and Pagerank into a single dataframe
centrality_df <- tibble(
  user = V(network_graph)$name,
  degree = central_degree,
  pagerank = pagerk
)
head(centrality_df)
```

```
# A tibble: 6 x 3
  user                degree pagerank
  <chr>                <dbl>    <dbl>
1 Silvia_Project           1  0.00142
2 novafuture               2  0.00215
3 nada_eid_88              1  0.00142
4 Hodhod_233               1  0.00142
5 bilalmohammedzeyada      1  0.00142
6 knowprose                2  0.00215
```

```r
# Top User by Degree Centrality
top_degree <- centrality_df |>
  arrange(desc(degree)) |>
  slice(1)

paste0("Top user by Degree Centrality:")
```

```
[1] "Top user by Degree Centrality:"
```

```r
print(top_degree)
```

```
# A tibble: 1 x 3
  user      degree pagerank
  <chr>      <dbl>    <dbl>
1 j_hearsay    134   0.0695
```

```r
# Top User by PageRank
top_pgrk <- centrality_df |>
  arrange(desc(pagerank)) |>
  slice(1)

paste0("Top user by PageRank:")
```

```
[1] "Top user by PageRank:"
```

```r
print(top_pgrk)
```

```
# A tibble: 1 x 3
  user         degree pagerank
  <chr>         <dbl>    <dbl>
1 artificial3d    116    0.129
```

Evaluation on the top user result of Degree Centrality and PageRank

The figures indicate a clear differentiation between popularity and influence within the network structure. J_hearsay user exhibits the highest Degree Centrality, with 134 connections. This indicates that this is the most active account, possessing the broadest reach in discussions about AI. In contrast, artificial3d user achieves the highest PageRank score (0.129), despite having fewer direct connections (116) than j_hearsay (134).

As a result, the disparity suggests that, although j_hearsay has a greater number of connections, artificial3d's connections are superior in terms of quality owing to influential followers. From a network perspective, artificial3d is a more reputable account whose credibility is validated by followers having high-scoring PageRank within the AI community, rather than solely by its raw number of connections.